IEEE Standards for LAN

IEEE 802.3 | IEEE 802.4 | IEEE 802.5

This document provides a detailed explanation of three foundational IEEE 802 Local Area Network (LAN) standards — covering their topology, access methods, key features, and real-world applications.

What is IEEE 802?

IEEE (Institute of Electrical and Electronics Engineers) launched the **802 Project** to define open standards for Local Area Networks (LANs). These standards specify how data is transmitted over a shared network medium — covering both physical connections and data link layer protocols.

IEEE 802.3 — Ethernet (CSMA/CD)

What is it?

IEEE 802.3 defines the standard for **Ethernet** — the most widely used LAN technology in the world today. It uses a method called **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)** to control how devices share the network.

Physical Setup

- Originally used a **Bus topology** — all devices on one shared cable.
- Later evolved to **Star topology** using hubs and switches.
- Cable types: **Coaxial** → **Twisted Pair (UTP)** → **Fiber Optic**

How CSMA/CD Works

Think of it like a group conversation at a dinner table — everyone shares the same channel and must take turns:

- **Carrier Sense** — Before transmitting, a device listens to check if the channel is free.
- **Multiple Access** — All devices share the same communication channel.
- **Collision Detection** — If two devices transmit simultaneously, a collision is detected; both stop immediately.
- **Backoff** — Each device waits a **random time** before retrying transmission.

Key Features

Feature	Detail
Topology	Bus (original) → Star (modern)
Access Method	CSMA/CD
Speed	10 Mbps → 100 Mbps → 1 Gbps → 10 Gbps+
Cable	Coaxial → Twisted Pair → Fiber
Frame Size	64 to 1518 bytes

■ ***Note:*** *The LAN cable (RJ-45) you plug into your laptop is an Ethernet cable — this is IEEE 802.3 in action and it is still the most widely used LAN standard today.*

IEEE 802.4 — Token Bus

What is it?

IEEE 802.4 defines the standard for **Token Bus** LAN. It combines the **physical structure of a Bus** network with a **logical ring** for token-based access control. It uses **coaxial broadband cable**.

How Token Passing Works

Think of a **'talking stick'** passed around in a meeting — only the person holding it may speak:

- A special signal called a **Token** circulates among all devices in a logical ring order.
- Only the device that **holds the token** can transmit data.
- After transmitting, the device **passes the token** to the next device in sequence.
- If a device has nothing to send, it immediately **passes the token** along.

Key Features

Feature	Detail
Topology	Physical Bus + Logical Ring
Access Method	Token Passing
Speed	1, 5, or 10 Mbps
Cable	Coaxial (Broadband)
Collision Possible?	No — only token holder transmits

Advantage over IEEE 802.3

Since only the token holder transmits, there are **no collisions**. This makes Token Bus **deterministic** — every device is guaranteed a turn within a known time frame, which is critical in time-sensitive environments.

■ ***Note:*** *Token Bus was widely used in* ***industrial automation and manufacturing*** *(e.g., factory floors). It is now largely* ***obsolete****, replaced by modern Ethernet.*

IEEE 802.5 — Token Ring

What is it?

IEEE 802.5 defines the standard for **Token Ring** LAN, originally developed by **IBM**. It uses a **ring topology** with the same token-passing access method, where a special frame (the token) circulates around the ring continuously.

Physical Setup

- Devices connected in a **Ring topology** — each device connects to exactly two neighbors.
- Uses a **MAU (Multistation Access Unit)** — looks like star physically, behaves like ring logically.
- Cable: **Shielded Twisted Pair (STP)**

How It Works (Step by Step)

- A **free Token** (3-byte special frame) continuously travels around the ring.
- A device wanting to send data waits to **capture the free token**.
- It marks the token as **'busy'**, attaches its data, and sends it around the ring.
- The **destination device** copies the data and marks the frame as 'received'.
- The frame returns to the **sender**, who releases a new free token for the next device.

Key Features

Feature	Detail
Topology	Physical Star / Logical Ring
Access Method	Token Passing
Speed	4 Mbps (original) → 16 Mbps
Cable	Shielded Twisted Pair (STP)
Collision Possible?	No
Priority System	Yes — 8 levels (0 to 7)

Priority Mechanism

Token Ring supports a **priority system from 0 to 7**. A device with a higher priority can **reserve the token** before a lower-priority device gets it. This makes it ideal for time-sensitive and real-time applications.

Advantage over IEEE 802.3

Token Ring is **fair and efficient** under heavy traffic — every device is guaranteed a turn. Ethernet (802.3) can suffer from repeated collisions under heavy network load, degrading performance.

■ ***Note:*** *Token Ring was popular in* ***IBM mainframe environments*** *in the 1980s–90s. It is now* ***obsolete*** *and has been completely replaced by Gigabit Ethernet.*

Quick Comparison: 802.3 vs 802.4 vs 802.5

Feature	802.3 (Ethernet)	802.4 (Token Bus)	802.5 (Token Ring)
Topology	Bus / Star	Bus (Logical Ring)	Ring (Logical Ring)
Access Method	CSMA/CD	Token Passing	Token Passing
Collision?	Yes	No	No
Speed	10 Mbps–10 Gbps+	1–10 Mbps	4–16 Mbps
Cable	Coaxial/UTP/Fiber	Coaxial	STP
Deterministic?	No	Yes	Yes
Priority System?	No	No	Yes (0–7)
Current Status	Widely Used	Obsolete	Obsolete
Best For	**General LAN**	Industrial/Factory	IBM Environments

Key Takeaways

- **IEEE 802.3 (Ethernet)** → Uses CSMA/CD, Bus/Star topology, still the dominant LAN standard worldwide.
- **IEEE 802.4 (Token Bus)** → Combines physical bus with logical ring; collision-free, used in industrial settings.
- **IEEE 802.5 (Token Ring)** → IBM's ring-based standard; collision-free with priority support, now obsolete.

The fundamental difference lies in the **access control method**: IEEE 802.3 uses **collision-based (CSMA/CD)** access, while IEEE 802.4 and 802.5 both use **token-based (collision-free)** access. Today, Ethernet (802.3) has evolved to gigabit speeds and dominates all modern LANs.